PHYSICAL CHEMISTRY CUMULATIVE EXAM

October 3, 1992

PART A: A 50.0 gram sample of pure water in equilibrium with its surroundings at 25°C is placed in a freezer at -18°C until it becomes ice at -18°C. First, draw and label a simple cooling curve for this process; what are the entropy changes incurred for the system, the surroundings, and the universe at each stage of the cooling?

[The heat capacities are C_p(ice) = 38.2, C_p(water) = 75.5 J K^{-1} mol^{-1}; the heat of fusion for H_2O is 6.008 kJ mol^{-1}]

PART B: The 50.0 gram ice chunk is now removed from the freezer compartment and returned to its original surroundings. Sometime later, the ice has completely melted and thermal equilibrium reestablished. First, draw and label a simple heating curve for this process; using the data in Part A above, what are the entropy changes for the system, the surroundings, and the universe at each stage of the heating?